Experimental and method

LOGICLAB KIDS — EOCYS 2026

Native Platforms

01

Native **SwiftUI (MVVM)** on iOS (live on Apple TestFlight) and **Kotlin Compose** on Android. Operates offline-first; all lessons execute with zero network connectivity.

Stage Grammar

02

6 modules and 36 lessons bundled as JSON. 4 interactive task types: Quiz, Spot-It, Sort-It, Build-It. Mistakes are never punished (explanations shown after 2 misses).

Premack Contingency

03

Applies the Premack (1959) differential probability law. Solving a lesson credits the ledger with **+20 pts & +5 min**. Child spends balance within parent-allowed hours.

Constrained Gemini

04

System prompt restricts Gemini to Socratic dialogue: probes reasoning and gives hints, never direct answers. Strict low token cap prevents lecturing; unlocks only post-task.

Interactive Child Experience · iOS & Android

01 — DAILY MYSTERY

02 — EARN FREE TIME

The Behavioral Lock

Screen entertainment is the highest-probability behavior, structured reasoning practice is among the lowest. The lock and key are in the OS: a child cannot negotiate around it.

The Parent's Side

Governed, not policed

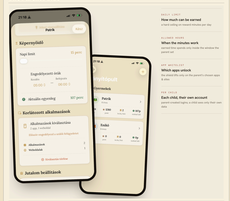

FIG 2. The parent sets the economy once — daily ceiling, allowed window, whitelisted apps, one account per child. Everything after that is enforced by iOS FamilyControls, not by the app. Screens: stripped iOS build.

Opaque Token Shielding

The system picker returns **opaque tokens**, so LogicLab Kids never accesses which apps the parent selected. Shields are applied directly by iOS ManagedSettings, not in-app.

Inside a Lesson

Story → Task → Explanation

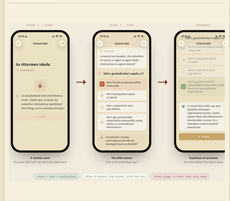

FIG 4. The "Slippery slope" lesson: the concept arrives inside a scene the child knows, becomes a task the child must solve, and a wrong answer ends in an explanation — never a punishment. The AI tutor comes last, never first.

4 Core Task Types

Quiz (sound confirmation), **Spot-It** (faulty step), **Sort-It** (inference chain), and **Build-It** (conclusion assembly). The child acts dynamically rather than memorizing definitions.

36	5 min	100%	0 Net
Interactive Lessons	**Earned Screen Time**	**OS-Level Shielding**	**Offline Runtime**
6 modules, across 4 tracks	Of play per earned lesson	Apple FamilyControls & Device Tokens	Zero telemetry required for lessons

REFERENCES

[1] Apple Inc. FamilyControls, ManagedSettings & DeviceActivity Frameworks.

[2] Premack, D. (1959). Toward empirical behavior laws I: Positive reinforcement.

[3] Kosmyna et al. (2025). Your Brain on ChatGPT. MIT Media Lab.

[4] LogicLab Kids System Architecture & Build Specification (2026).